SBT-DF204 COMPUTER FORENSICS CASE STUDY 1

INVESTIGATING HARASSMENT EMAIL TRAFFIC WITH WIRESHARK

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Registration No.: 2025/FWSD/11466
Course: SBT-DF204: Computer Forensics Case Studies
Analysis Platform: Kali Linux, Wireshark 4.6.6, TShark 4.6.6
Evidence Examined: `nitroba.pcap`
Case Study: Investigating Harassment Email Traffic With Wireshark

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Investigating Harassment Email Traffic With Wireshark

1. EXECUTIVE SUMMARY

This forensic investigation examined the supplied `nitroba.pcap` network capture to determine whether a client device accessed the WillSelfDestruct web service and transmitted the documented harassment message.

The investigation identified client IP address **192.168.15.4** communicating with service IP address **69.25.94.22**. Packet **82936** records an HTTP GET request to `www.willselfdestruct.com/secure/submit`, while packet **83601** records an HTTP POST request to the same endpoint containing the recipient, subject and message form fields.

The POST request identified the recipient as lilytuckrige@yahoo.com and the subject as "**you can't find u**s". Following the TCP stream associated with packet 83601 confirmed the HTTP submission and showed the server responding with an HTTP 302 redirect to `/secure/success`.

The relevant traffic was associated with source MAC address **00:17:f2:e2:c0:ce**

Additional browser/session evidence was identified in packet **78967**. Traffic from the same client contained a Gmail cookie associated with jcoachj@gmail.com . This provides technical browser/session evidence but does not, by itself, prove the legal or physical identity of the person who controlled the session.

The evidence therefore provides **high confidence that the observed client/device context generated the documented harassment-message submission**, but **lower confidence for attribution to a specific individual** . IP addresses and MAC addresses identify network endpoints or interfaces rather than people, and the case scenario involves shared/open wireless access. The Chem 109 roster was also not available in the local laboratory evidence, so no roster based identity conclusion is asserted.

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2. CASE INFORMATION AND SCOPE

2.1 Case Identification

Field	Details

Student	Charles Uke
Registration No.	2025/FWSD/11466
Course	SBT-DF204: Computer Forensics Case Studies
Case Study	Investigating Harassment Email Traffic With Wireshark
Evidence	`nitroba.pcap`
Analysis Platform	Kali Linux
Wireshark	4.6.6
TShark	4.6.6
Primary Client IP	192.168.15.4
Primary Service IP	69.25.94.22
Web Service	www.willselfdestruct.com

2.2 Investigation Objectives

The investigation was conducted to:

1. Identify and preserve the supplied forensic evidence.
2. Verify evidence integrity using cryptographic hashing.
3. Identify the client that contacted the relevant web service.
4. Identify the web service and relevant HTTP requests.
5. Correlate the client with the harassment-message submission.
6. Identify the MAC address associated with the relevant client traffic.
7. Examine browser/session evidence for possible account correlation.
8. Establish the exact network-event timeline.
9. Assess whether the available evidence supports person-level attribution.
10. Document limitations and alternative explanations.

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3. EVIDENCE ACQUISITION AND PRESERVATION

The supplied assessment materials included an archive named:

`nitroba.zip`

The archive contained the assessment PCAP:

`nitroba.pcap`

The original archive and extracted PCAP were preserved separately. A forensic working copy was then created so that analysis could be performed without modifying the preserved evidence.

3.1 Original Archive

Filename: `nitroba.zip`
Approximate size: 38,528 KB

SHA-256:

`d9685337c2a76129c038bd769afda473a5968048f874c30432f8e0d44360fd81`

3.2 Extracted Evidence

Filename: `nitroba.pcap`
Size: 56,180,821 bytes

SHA-256:

`2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb`

3.3 Working Copy

Filename:

`nitroba_working.pcap`

Location:

`~/SBT-DF204-CaseStudy1/working/`

SHA-256:

`2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb`

The extracted evidence and working copy produced the **same SHA-256 hash**.

This demonstrates that the working copy was identical to the preserved evidence at the time of verification.

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4. PCAP CHARACTERISTICS

The supplied PCAP was examined using capture-analysis utilities and Wireshark/TShark.

The capture characteristics were recorded as follows:

* Capture format: PCAP
* Timestamp precision: microsecond
* Byte order: little-endian
* PCAP version: 2.4
* Link type: Ethernet
* Capture length: 4096 bytes
* Approximate packet count: 94,000
* Data size: approximately 54 MB
* Capture duration: 15,759.950751 seconds
* Earliest observed packet: `2008-07-21 21:51:07.095278`
* Latest observed packet: `2008-07-22 02:13:47.046029`
* Average packet size: 579.07 bytes
* Average packet rate: approximately 5 packets/second
* Strict packet time order: true
* Interface count: one Ethernet interface

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5. FORENSIC METHODOLOGY

The investigation followed a preservation-first forensic workflow.

The methodology consisted of:

1. Identifying the supplied evidence.
2. Calculating SHA-256 hashes.
3. Preserving the original evidence.

4. Creating a separate working copy.
5. Verifying the working copy hash.
6. Examining PCAP characteristics.
7. Using targeted Wireshark/TShark display filters.
8. Identifying HTTP requests to the relevant service.
9. Identifying the relevant HTTP POST transaction.
10. Examining HTTP form fields.
11. Following the associated TCP stream.
12. Correlating IP and MAC addresses.
13. Examining browser/session evidence.
14. Reconstructing the event timeline.
15. Searching for Chem 109 evidence.
16. Separating direct observations from reasonable inferences.
17. Assessing attribution limitations.
18. Producing a final defensible conclusion.

No person-level conclusion was based solely on an IP address, MAC address, or browser/account identifier.

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6. DETAILED FORENSIC FINDINGS

6.1 What File Was Acquired and Preserved?

The assessment evidence was preserved as:

`nitroba.pcap`

The original archive was:

`nitroba.zip`

The extracted PCAP SHA-256 was:

`2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb`

The forensic working copy produced the exact same SHA-256 hash.

Therefore, the evidence used for analysis was verified as identical to the preserved source.

6.2 Which Client Contacted the Web Service?

The relevant client was:

IP: `192.168.15.4`

The relevant service was:

IP: `69.25.94.22`

The service hostname was:

www.willselfdestruct.com

Packet **82936** recorded:

Timestamp:

`2008-07-22 06:03:43.825871 UTC`

Request:

`GET /secure/submit HTTP/1.1`

The request therefore establishes that client `192.168.15.4` accessed the WillSelfDestruct message-submission page.

6.3 What Links the Client to the Harassment Message?

The strongest evidence is packet **83601**

Packet 83601 recorded:

Timestamp:

`2008-07-22 06:04:24.311700 UTC`

Source:

`192.168.15.4`

Destination:

`69.25.94.22`

HTTP method:

`POST`

Request:

`POST /secure/submit HTTP/1.1`

Content-Type:

`application/x-www-form-urlencoded`

Content-Length:

`188`

The submitted form contained the following material fields:

Recipient:

`lilytuckrige@yahoo.com`

Subject:

`you can't find us`

The POST also contained the message field and additional form parameters:

* `type=0`
* `ttl=30`
* `submit.x=92`
* `submit.y=26`

The full hostile message is not reproduced in this report because the forensic conclusion does not require reproducing its complete content. The original packet remains the evidentiary source.

The HTTP POST therefore provides the strongest direct evidence linking the observed client to the transmission of the documented message.

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7. FOLLOW TCP STREAM ANALYSIS

The TCP stream associated with packet 83601 was followed in Wireshark.

The stream showed the HTTP POST request and the corresponding server response.

The server responded with:

`HTTP/1.1 302 Moved Temporarily`

The response also contained a redirect to:

`/secure/success`

This is significant because the stream-level evidence connects:

Client → submission page → POST containing message → server response → success page**

The sequence provides strong confirmation that the observed client submitted the information to the WillSelfDestruct service.

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8. DEVICE AND MAC ADDRESS ANALYSIS

The relevant source MAC address was:

`00:17:f2:e2:c0:ce`

The MAC address was associated with:

`192.168.15.4`

The Ethernet level evidence therefore provides device/interface correlation for the HTTP transaction.

However, the MAC address does not independently establish who was operating the device.

A MAC address identifies a network interface or device within the observed network environment. It does not inherently identify a human operator.

Therefore, the finding is:

> The device/interface associated with MAC `00:17:f2:e2:c0:ce` generated the relevant client traffic.

It is **not**:

> The MAC address proves that a particular person operated the device.

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9. GMAIL SESSION CORRELATION

Packet **78967** provided additional browser/session evidence.

The traffic originated from:

`192.168.15.4`

The packet involved Gmail infrastructure.

The HTTP cookie included:

`gmailchat=jcoachj@gmail.com/475090`

This provides technical evidence that a browser/session associated with the Gmail account identifier:

`jcoachj@gmail.com`

was active from the same client context.

This evidence should be interpreted carefully.

It establishes:

* browser/session association;
* account identifier presence;
* technical correlation with the same client context.

It does *not* independently establish:

* the legal identity of the account holder;
* the physical identity of the person operating the computer;
* that the account holder personally submitted the harassment message.

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10. CHEM 109 ROSTER INVESTIGATION

The investigation attempted to determine whether the relevant account/person could be correlated with a Chem 109 roster.

A local search was performed for potentially relevant documents, including:

* PDF files;
* PowerPoint files;
* Word documents;
* text files;

* HTML files.

The available local desktop material did not contain the Chem 109 roster.

A TShark search for the exact phrase:

`Chem 109`

returned no relevant direct roster evidence in the captured frames.

A broader search for:

`chem`

produced numerous unrelated results and was not suitable for establishing identity.

Consequently, no roster-based identity conclusion is made.

The appropriate forensic statement is:

> The Chem 109 roster was not available in the local laboratory evidence supplied for this investigation; therefore, no roster-based identification is asserted.

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11. TIMESTAMP AND EVENT ANALYSIS

The primary timeline uses the **PCAP frame timestamps**

11.1 Gmail Session

Packet: 78967

Time:

`2008-07-22 06:01:02.113436 UTC`

The same client generated Gmail/session traffic containing the observed Gmail cookie.

11.2 WillSelfDestruct Submission Page

Packet: 82936

Time:

`2008-07-22 06:03:43.825871 UTC`

The client accessed:

`GET /secure/submit`

11.3 Page Resources

Between approximately:

`06:03:44 UTC`

the same client requested several resources associated with the submission page.

These included:

* `/images/spacer.gif`
* `/images/sm-logo.gif`
* `/images/warning-home.gif`
* `/images/body-bk.gif`
* `/images/bttn-send.gif`
* `/images/bridge_small.gif`

This activity is consistent with loading the submission page.

11.4 Message Submission

Packet: 83601

Time:

`2008-07-22 06:04:24.311700 UTC`

The client submitted:

`POST /secure/submit`

with the documented form fields.

11.5 Successful Response

Packet: 83614

Time:

`2008-07-22 06:04:24.564165 UTC`

The client requested:

`GET /secure/success`

11.6 Additional Page Resource

Packet: 83654

Time:

`2008-07-22 06:04:24.902157 UTC`

The client requested another page resource after the success response.

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12. TIMESTAMP DISCREPANCY NOTE

The PCAP frame timestamp for packet 83601 is:

`2008-07-22 06:04:24.311700 UTC`

However, the HTTP Date header observed within the TCP stream reported:

`Tue, 22 Jul 2008 07:24:45 GMT`

These timestamps originate from different sources.

The first is the network capture timestamp assigned to the packet.

The second is an application/server HTTP header.

Both values have therefore been preserved as observed rather than replacing one with the other.

For the forensic event timeline, the **PCAP packet timestamp is treated as the primary network-event timestamp**.

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13. CHRONOLOGICAL FORENSIC TIMELINE

Time (UTC)	Packet	Event
06:01:02.113436	78967	Gmail/session traffic from 192.168.15.4 containing the observed Gmail cookie
06:03:43.825871	82936	GET `/secure/submit` to www.willselfdestruct.com
06:03:44	82985–83162	Multiple HTTP requests for page resources
06:04:24.311700	83601	POST `/secure/submit` containing recipient, subject and message fields
06:04:24.564165	83614	GET `/secure/success`
06:04:24.902157	83654	Additional post-submission page-resource request

This sequence provides a coherent network-level reconstruction of the submission event.

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14. DIRECT OBSERVATIONS

The following findings were directly observed in the evidence:

1. The investigation examined `nitroba.pcap`.
2. The extracted PCAP had SHA-256:
   `2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb`.
3. The working copy had the same SHA-256.
4. Client IP `192.168.15.4` accessed the WillSelfDestruct service.
5. Service IP `69.25.94.22` was contacted.
6. Packet 82936 contained `GET /secure/submit`.

7. Packet 83601 contained `POST /secure/submit`.
8. Packet 83601 contained recipient, subject and message fields.
9. The recipient was `lilytuckrige@yahoo.com`.
10. The subject was `you can't find us`.
11. The source MAC was `00:17:f2:e2:c0:ce`.
12. Packet 83614 showed a request for `/secure/success`.
13. Packet 78967 contained Gmail/session evidence associated with `jcoachj@gmail.com`.
14. The Chem 109 roster was not available in the local evidence.

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15. REASONABLE FORENSIC INFERENCES

The evidence supports the following reasonable inferences:

1. The device/interface associated with MAC `00:17:f2:e2:c0:ce` generated the relevant traffic.
2. Client IP `192.168.15.4` accessed the WillSelfDestruct submission page.
3. The same client subsequently submitted the HTTP POST containing the documented harassment message.
4. The server processed the submission and redirected the client to `/secure/success`.
5. The Gmail/session evidence is technically correlated with the same client context.
6. The observed traffic forms a coherent chronological sequence.

These are technical/network inferences and should not be expanded into unsupported claims about the physical identity of a person.

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16. ATTRIBUTION LIMITATIONS

Several limitations affect person-level attribution.

16.1 IP Address Limitation

An IP address identifies a network endpoint during the observed traffic. It does not inherently identify the human using that endpoint.

16.2 MAC Address Limitation

A MAC address identifies a network interface/device within the captured environment. It does not prove who was physically operating the device.

16.3 Shared/Open Wireless Environment

The scenario involves shared/open wireless access.

This creates an important attribution limitation because other authorized or unauthorized users may potentially have had access to the same network environment.

16.4 Browser Cookie Limitation

The Gmail cookie provides evidence of an account/session association.

It does not independently prove:

* who owned the account;
* who controlled the browser;

* who physically operated the computer.

16.5 Missing Chem 109 Roster

The Chem 109 roster was not available in the local laboratory evidence.

Therefore, no attempt has been made to match the Gmail identifier to a roster entry.

16.6 Alternative Explanation

The technical evidence is consistent with the observed device generating the submission, but another individual could potentially have used the same device, account/session, or network connection.

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17. DEFENSIBLE CONCLUSION

Overall Technical Assessment

HIGH CONFIDENCE

The network evidence provides strong technical evidence that the client/device context at:

IP: `192.168.15.4`

associated with:

MAC: `00:17:f2:e2:c0:ce`

accessed:

www.willselfdestruct.com

and submitted the documented harassment message through:

`POST /secure/submit`

at approximately:

`2008-07-22 06:04:24.311700 UTC`

The HTTP POST, TCP stream, MAC address, subsequent `/secure/success` request and chronological client correlation provide mutually supporting evidence of the submission event.

Person-Level Attribution

LOWER CONFIDENCE

The evidence is insufficient to definitively identify the physical person who operated the device.

The IP address, MAC address and Gmail cookie provide strong technical correlation but do not independently establish human identity.

The shared/open wireless environment and absence of the Chem 109 roster further limit person level attribution.

Accordingly, the most defensible conclusion is:

> **There is high confidence that the observed client/device context generated the documented harassment message submission, but lower confidence that the available evidence alone can attribute that activity to a specific individual.**

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18. COMPLETE COMMAND RECORD

The following commands were used during the investigation and final verification.

Command Set 1 — Enter Case Workspace

```bash
cd ~/SBT-DF204-CaseStudy1
```

Command Set 2 — Create Forensic Working Copy

```bash
cp --preserve=mode,timestamps evidence/source/nitroba.pcap working/nitroba_working.pcap
```

Command Set 3 — Verify Evidence Hashes

```bash
sha256sum evidence/source/nitroba.pcap
sha256sum working/nitroba_working.pcap
```

Expected matching SHA-256:

```text
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
```

Command Set 4 — Identify File Type and Size

```bash
file evidence/source/nitroba.pcap

ls -lh evidence/source/nitroba.pcap

ls -lh working/nitroba_working.pcap
```

Command Set 5 — Examine PCAP Statistics

```bash
capinfos evidence/source/nitroba.pcap
```

Command Set 6 — Discover WillSelfDestruct HTTP GET Requests

```bash
tshark -r working/nitroba_working.pcap \
-Y 'http.request.method == "GET" && http.host == "www.willselfdestruct.com"'
```

```

This identified packet 82936 and related page-resource requests.

**Command Set 7 — Examine HTTP POST Requests**

```bash
tshark -r working/nitroba_working.pcap \
-Y 'http.request.method == "POST"'
```

This was used to locate HTTP POST activity within the capture.

Command Set 8 — Isolate the WillSelfDestruct POST

```bash
tshark -r working/nitroba_working.pcap \
-Y 'http.request.method == "POST" && http.host == "www.willselfdestruct.com"'
```

This identified:

Packet 83601

Command Set 9 — Correlate Client MAC With Service Traffic

```bash
tshark -r working/nitroba_working.pcap \
-Y 'eth.src == 00:17:f2:e2:c0:ce && ip.dst == 69.25.94.22 && http'
```

This produced the chronological sequence involving:

* packet 82936;
* page resources;
* packet 83601;
* packet 83614;
* packet 83654.

Command Set 10 — Search for Gmail Cookie Evidence

```bash
tshark -r working/nitroba_working.pcap \
-Y 'eth.src == 00:17:f2:e2:c0:ce && http.cookie contains "@"'
```

This identified packet 78967 and the observed Gmail cookie.

Command Set 11 — Search for Chem 109 Evidence

```bash
tshark -r working/nitroba_working.pcap \
-Y 'frame contains "Chem 109"'
```

Result:
```

No relevant matching frame was identified.

A broader search was also performed:

```bash
tshark -r working/nitroba_working.pcap \
-Y 'frame contains "chem"'
```

This produced numerous unrelated results and was therefore not treated as roster evidence.

Command Set 12 — Search for Gmail Account Identifier

```bash
tshark -r working/nitroba_working.pcap \
-Y 'frame contains "jcoachj@gmail.com"'
```

This produced multiple frames containing the account identifier.

Command Set 13 — Search Local Desktop for Roster Documents

```bash
find ~/Desktop -type f \
\( -iname "*.pdf" -o -iname "*.ppt" -o -iname "*.pptx" \
-o -iname "*.doc" -o -iname "*.docx" -o -iname "*.txt" \
-o -iname "*.html" \) -printf '%p\n' | sort
```

The available results did not contain a Chem 109 roster.

Command Set 14 — Final Evidence Integrity Verification

```bash
cd ~/SBT-DF204-CaseStudy1

sha256sum evidence/source/nitroba.pcap

sha256sum working/nitroba_working.pcap

ls -lh notes/evidence_log.txt

ls -lh notes/final_findings.txt

ls -lh notes/final_timeline.txt

find . -maxdepth 2 -type f | sort
```

The source and working-copy SHA-256 values matched.

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19. SCREENSHOT REGISTER

The following screenshots were confirmed as available for the investigation.

Screenshot	Description
SS-01	Evidence acquisition and original SHA-256
SS-02	Working-copy integrity
SS-03	Initial Wireshark packet view
SS-04	WillSelfDestruct web traffic discovery
SS-05	HTTP `/secure/submit` request
SS-06	Harassment-message HTTP POST
SS-07	Follow TCP Stream message submission
SS-08	Client MAC address evidence
SS-09	Gmail cookie/session evidence
SS-10	Harassment client traffic correlation
SS-11	Final forensic timeline

Screenshot Evidence Mapping

SS-01 supports evidence acquisition and integrity.

SS-02 supports working-copy verification.

SS-03 documents the initial Wireshark examination.

SS-04 supports discovery of the WillSelfDestruct service.

SS-05 supports the HTTP GET request to `/secure/submit`.

SS-06 supports the HTTP POST containing the message form data.

SS-07 supports the TCP-stream correlation and server response.

SS-08 supports the MAC/IP device correlation.

SS-09 supports Gmail/session correlation.

SS-10 supports the chronological client traffic sequence.

SS-11 supports the final forensic timeline.

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20. EVIDENCE LOG

Evidence ID	Packet/Item	Finding	Why It Matters
E-01	`nitroba.zip`	Original assessment archive and SHA-256 recorded	Acquisition/provenance
E-02	`nitroba.pcap`	Preserved extracted PCAP	Primary evidence
E-03	`nitroba_working.pcap`	Working copy with matching SHA-256	Analysis integrity
E-04	Packet 78967	Gmail/session cookie evidence	Session correlation
E-05	Packet 82936	GET `/secure/submit`	Service access
E-06	Packet 83601	POST `/secure/submit` with form fields	Direct message-submission evidence
E-07	TCP stream for 83601	POST plus HTTP 302 response	Stream correlation

E-08	Packet 83601 Ethernet/IP	MAC and client IP correlation	Device/interface correlation
E-09	Packet 83614	GET `/secure/success`	Submission sequence
E-10	Final timeline	Chronological reconstruction	Timeline analysis

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21. FORENSIC INTEGRITY STATEMENT

The investigation was conducted using a preserved evidence source and a separate working copy.

The SHA-256 values were independently checked and found to match:

Evidence source:

`2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb`

Working copy:

`2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb`

This provides evidence that the working copy used during analysis corresponded to the preserved source.

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22. FINAL STATEMENT

This report documents the forensic examination of the supplied `nitroba.pcap` evidence using Wireshark and TShark.

The strongest evidence establishes that the client/device context associated with **192.168.15.4** and **MAC address 00:17:f2:e2:c0:ce** accessed the WillSelfDestruct service and submitted the documented harassment message.

The HTTP GET, HTTP POST, form fields, TCP stream, MAC address, Gmail/session evidence and subsequent success request provide mutually supporting technical evidence.

However, forensic attribution must remain appropriately qualified. The evidence does not independently establish the physical identity of the person who operated the device. The shared/open wireless environment, the limitations of IP/MAC/account identifiers, and the absence of the Chem 109 roster prevent a definitive person-level identification.

Final assessment:

HIGH CONFIDENCE — The observed client/device context generated the documented harassment-message submission.

LOWER CONFIDENCE — Attribution of that activity to a specific human individual.

END OF REPORT

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SBT-DF204: Computer Forensics Case Studies

```
┌──(kali㉿kali)-[~/Desktop/Investigating Harassment Email Traffic With Wireshark]
└─$ cd ~/SBT-DF204-CaseStudy1

mkdir -p evidence/source

unzip -p ~/Desktop/"Investigating Harassment Email Traffic With Wireshark"/nitroba.zip \
> evidence/source/nitroba.pcap

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ ls -lh evidence/source/nitroba.pcap

file evidence/source/nitroba.pcap

sha256sum evidence/source/nitroba.pcap

capinfos evidence/source/nitroba.pcap | head -25
-rw-rw-r-- 1 kali kali 54M Oct  4 03:31 evidence/source/nitroba.pcap
evidence/source/nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/source/nitroba.pcap
File name:           evidence/source/nitroba.pcap
File type:           Wireshark/tcpdump/ ...  - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 4096 bytes
Number of packets:   94 k
File size:           56 MB
Data size:           54 MB
Capture duration:    15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time:   2008-07-22 02:13:47.046029
Data byte rate:      3,468 bytes/s
```

```
sha256sum evidence/source/nitroba.pcap

capinfos evidence/source/nitroba.pcap | head -25
-rw-rw-r-- 1 kali kali 54M Oct  4 03:31 evidence/source/nitroba.pcap
evidence/source/nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/source/nitroba.pcap
File name:           evidence/source/nitroba.pcap
File type:           Wireshark/tcpdump/ ...  - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 4096 bytes
Number of packets:   94 k
File size:           56 MB
Data size:           54 MB
Capture duration:    15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time:   2008-07-22 02:13:47.046029
Data byte rate:      3,468 bytes/s
Data bit rate:       27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256:              2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1:                65656392412add15f93f8585197a8998aaeb50a1
Strict time order:   True
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = Ethernet (1 - ether)
                     Capture length = 4096
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0
```

```
                    Encapsulation = Ethernet (1 - ether)
                    Capture length = 4096
                    Time precision = microseconds (6)
                    Time ticks per second = 1000000
                    Number of stat entries = 0

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ cd ~/SBT-DF204-CaseStudy1

mkdir -p evidence/source

unzip -p ~/Desktop/"Investigating Harassment Email Traffic With Wireshark"/nitroba.zip \
> evidence/source/nitroba.pcap

ls -lh evidence/source/nitroba.pcap

file evidence/source/nitroba.pcap

sha256sum evidence/source/nitroba.pcap

capinfos evidence/source/nitroba.pcap | head -25
-rw-rw-r-- 1 kali kali 54M Oct  4 03:33 evidence/source/nitroba.pcap
evidence/source/nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/source/nitroba.pcap
File name:           evidence/source/nitroba.pcap
File type:           Wireshark/tcpdump/ ... - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 4096 bytes
Number of packets:   94 k
File size:           56 MB
Data size:           54 MB
```

```
└─$ cd ~/SBT-DF204-CaseStudy1

mkdir -p evidence/source

unzip -p ~/Desktop/"Investigating Harassment Email Traffic With Wireshark"/nitroba.zip \
> evidence/source/nitroba.pcap

ls -lh evidence/source/nitroba.pcap

file evidence/source/nitroba.pcap

sha256sum evidence/source/nitroba.pcap

capinfos evidence/source/nitroba.pcap | head -25
-rw-rw-r-- 1 kali kali 54M Oct  4 03:33 evidence/source/nitroba.pcap
evidence/source/nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/source/nitroba.pcap
File name:           evidence/source/nitroba.pcap
File type:           Wireshark/tcpdump/ ...  - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 4096 bytes
Number of packets:   94 k
File size:           56 MB
Data size:           54 MB
Capture duration:    15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time:   2008-07-22 02:13:47.046029
Data byte rate:      3,468 bytes/s
Data bit rate:       27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
```

```
capinfos evidence/source/nitroba.pcap | head -25
-rw-rw-r-- 1 kali kali 54M Oct  4 03:33 evidence/source/nitroba.pcap
evidence/source/nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/source/nitroba.pcap
File name:           evidence/source/nitroba.pcap
File type:           Wireshark/tcpdump/ ...  - pcap
File encapsulation:  Ethernet
File timestamp precision:  microseconds (6)
Packet size limit:   file hdr: 4096 bytes
Number of packets:   94 k
File size:           56 MB
Data size:           54 MB
Capture duration:    15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time:   2008-07-22 02:13:47.046029
Data byte rate:      3,468 bytes/s
Data bit rate:       27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256:              2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1:                65656392412add15f93f8585197a8998aaeb50a1
Strict time order:   True
Number of interfaces in file: 1
Interface #0 info:
                     Encapsulation = Ethernet (1 - ether)
                     Capture length = 4096
                     Time precision = microseconds (6)
                     Time ticks per second = 1000000
                     Number of stat entries = 0

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ cd ~/SBT-DF204-CaseStudy1

cp --preserve=mode,timestamps \
evidence/source/nitroba.pcap \
working/nitroba_working.pcap

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$ echo "═══ SOURCE EVIDENCE ═══"
sha256sum evidence/source/nitroba.pcap

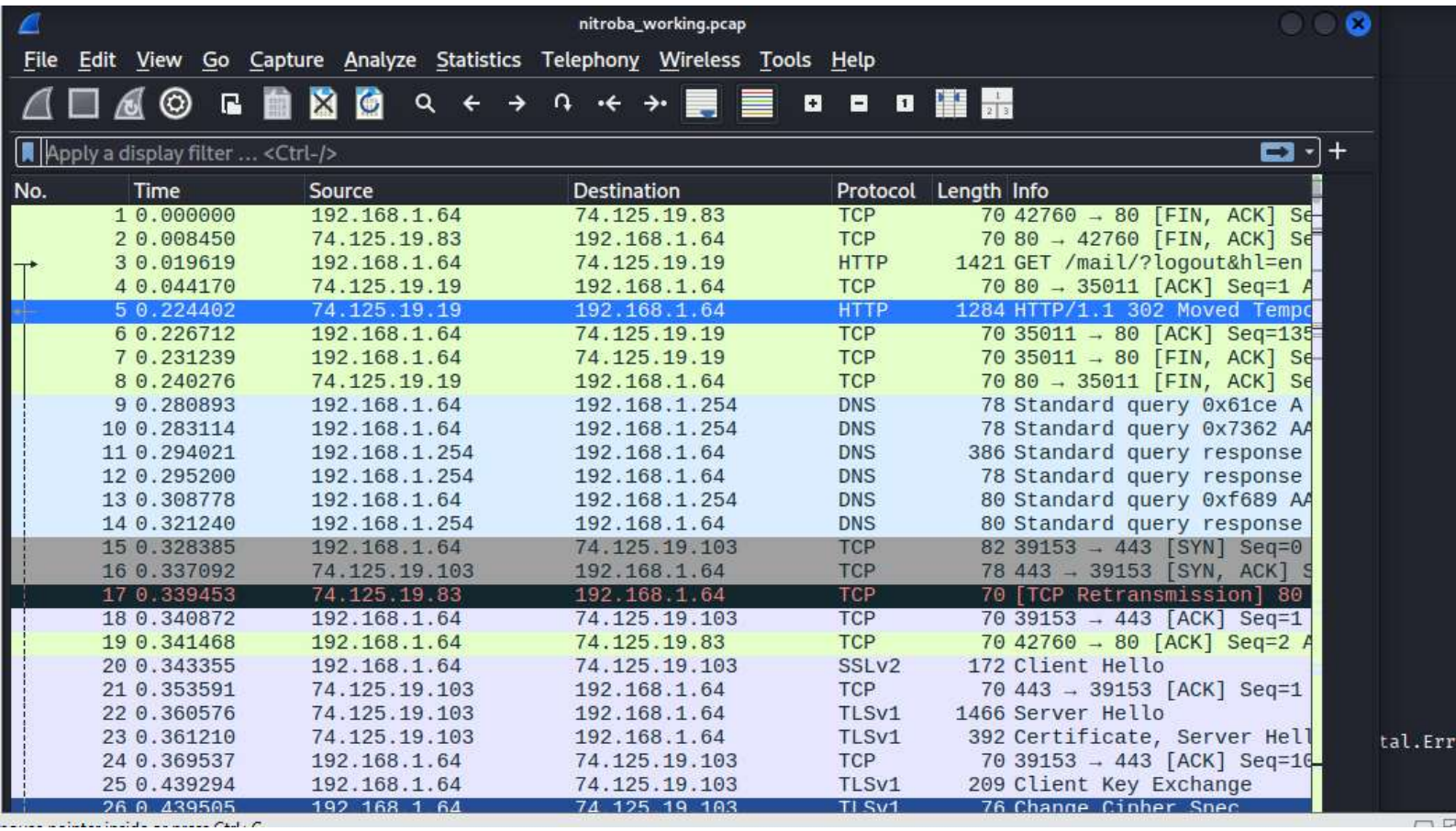
echo
echo "═══ WORKING COPY ═══"
sha256sum working/nitroba_working.pcap

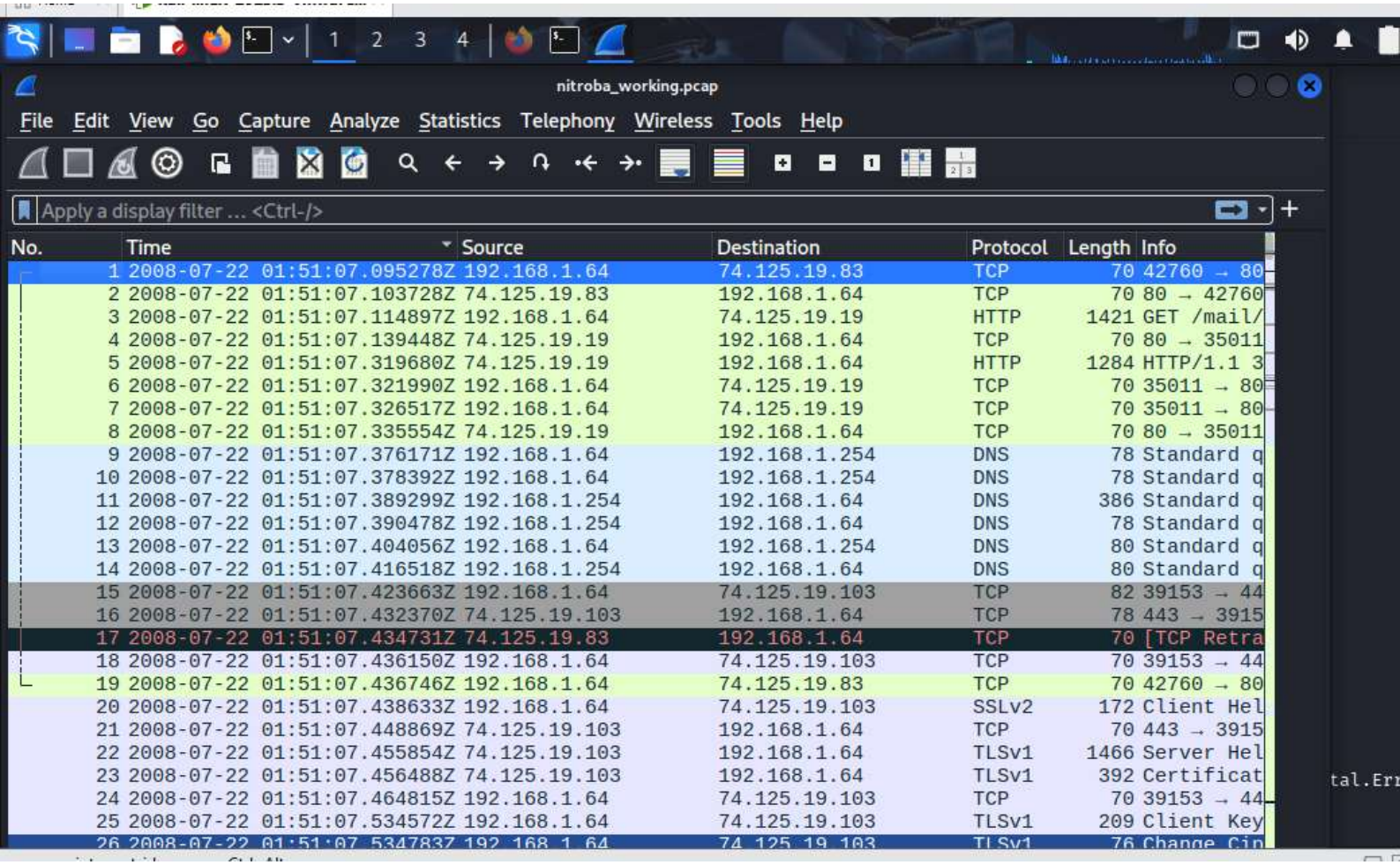
echo
echo "═══ FILES ═══"
ls -lh evidence/source/nitroba.pcap working/nitroba_working.pcap
═══ SOURCE EVIDENCE ═══
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  evidence/source/nitroba.pcap

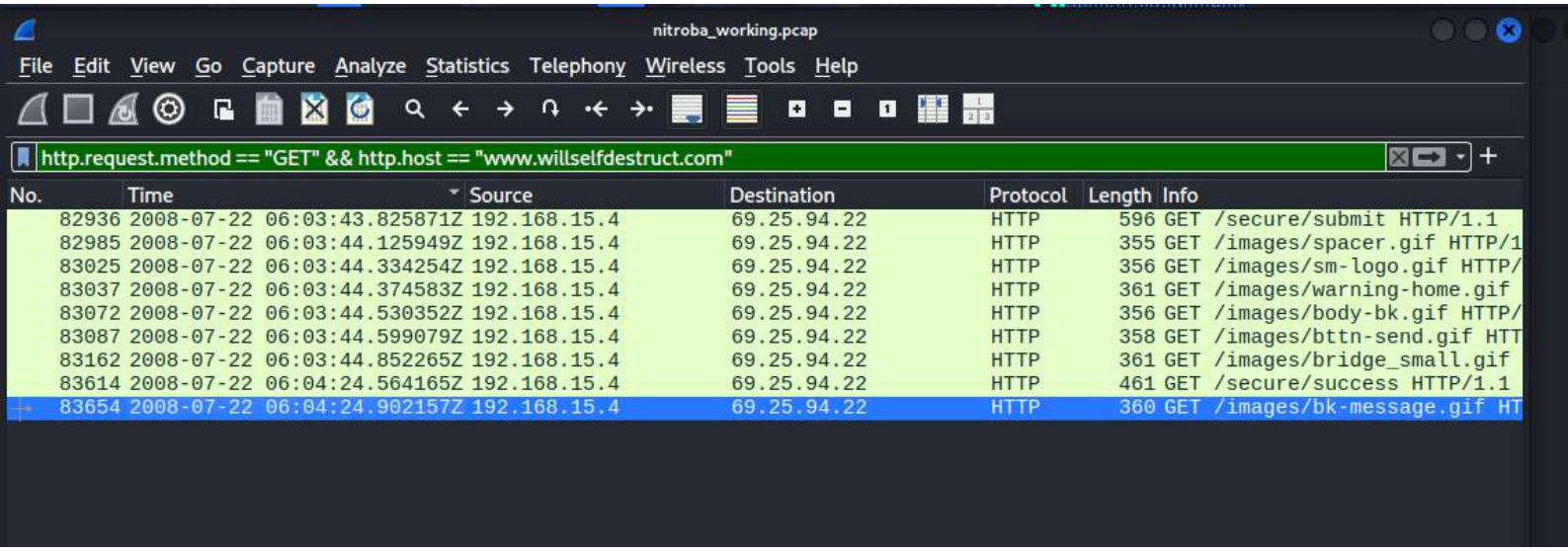
═══ WORKING COPY ═══
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb  working/nitroba_working.pcap

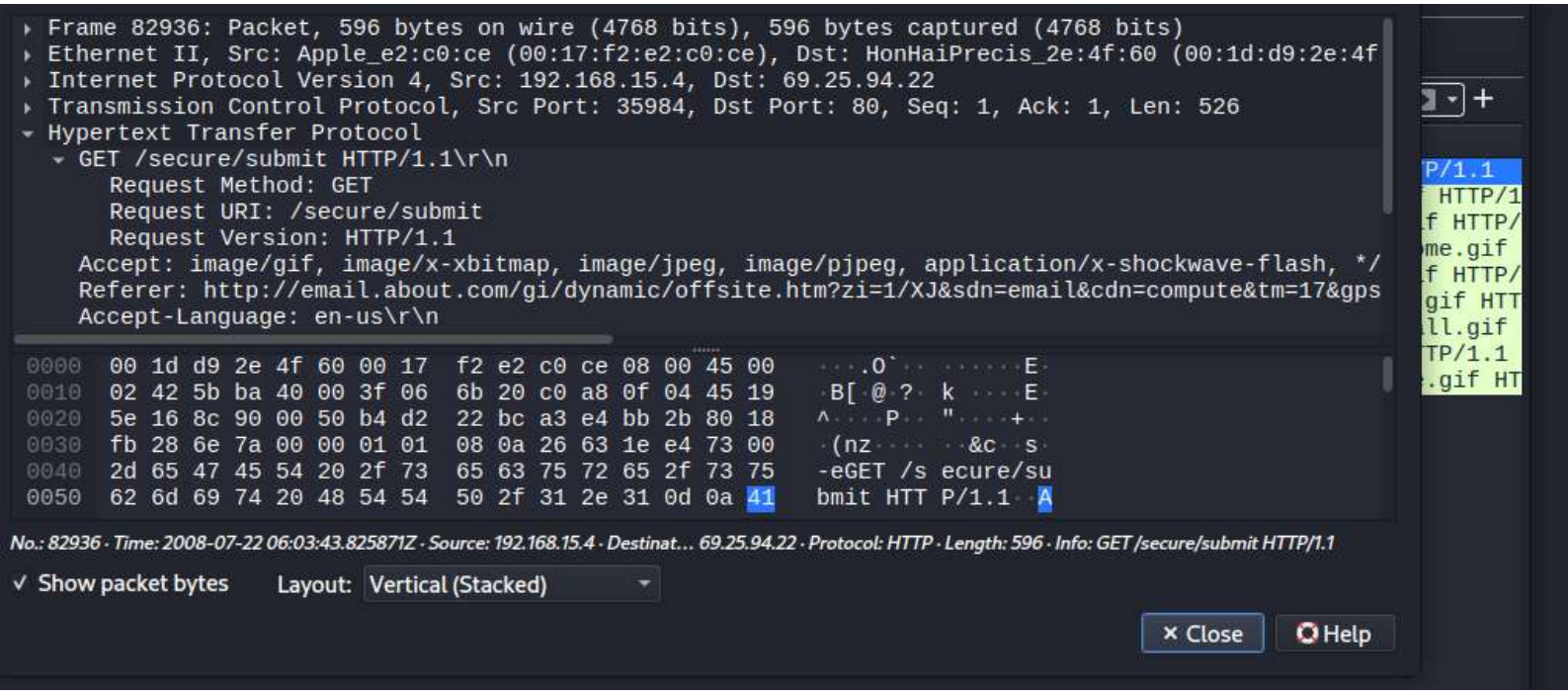
═══ FILES ═══
-rw-rw-r-- 1 kali kali 54M Oct  4 03:33 evidence/source/nitroba.pcap
-rw-rw-r-- 1 kali kali 54M Oct  4 03:33 working/nitroba_working.pcap

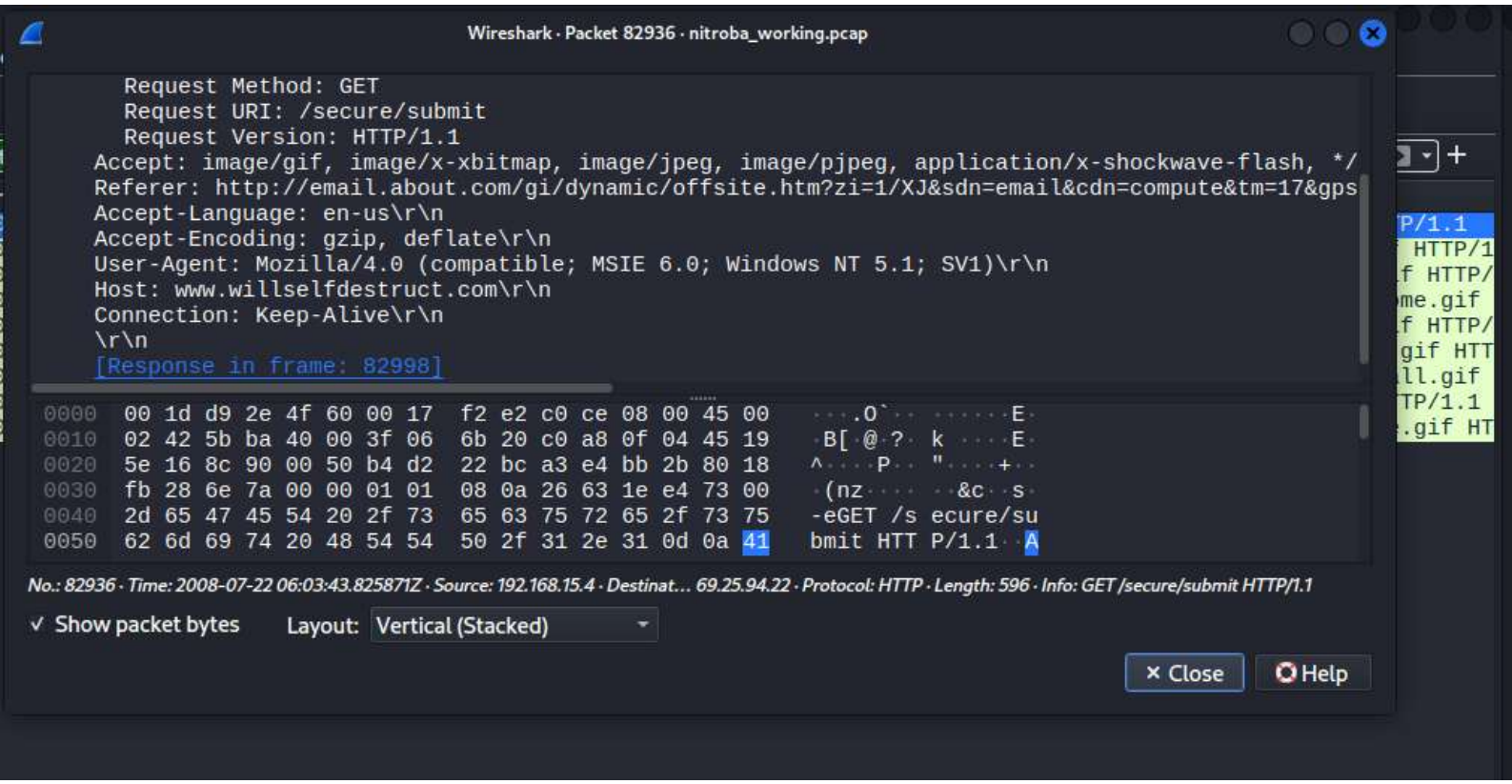
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy1]
└─$
```

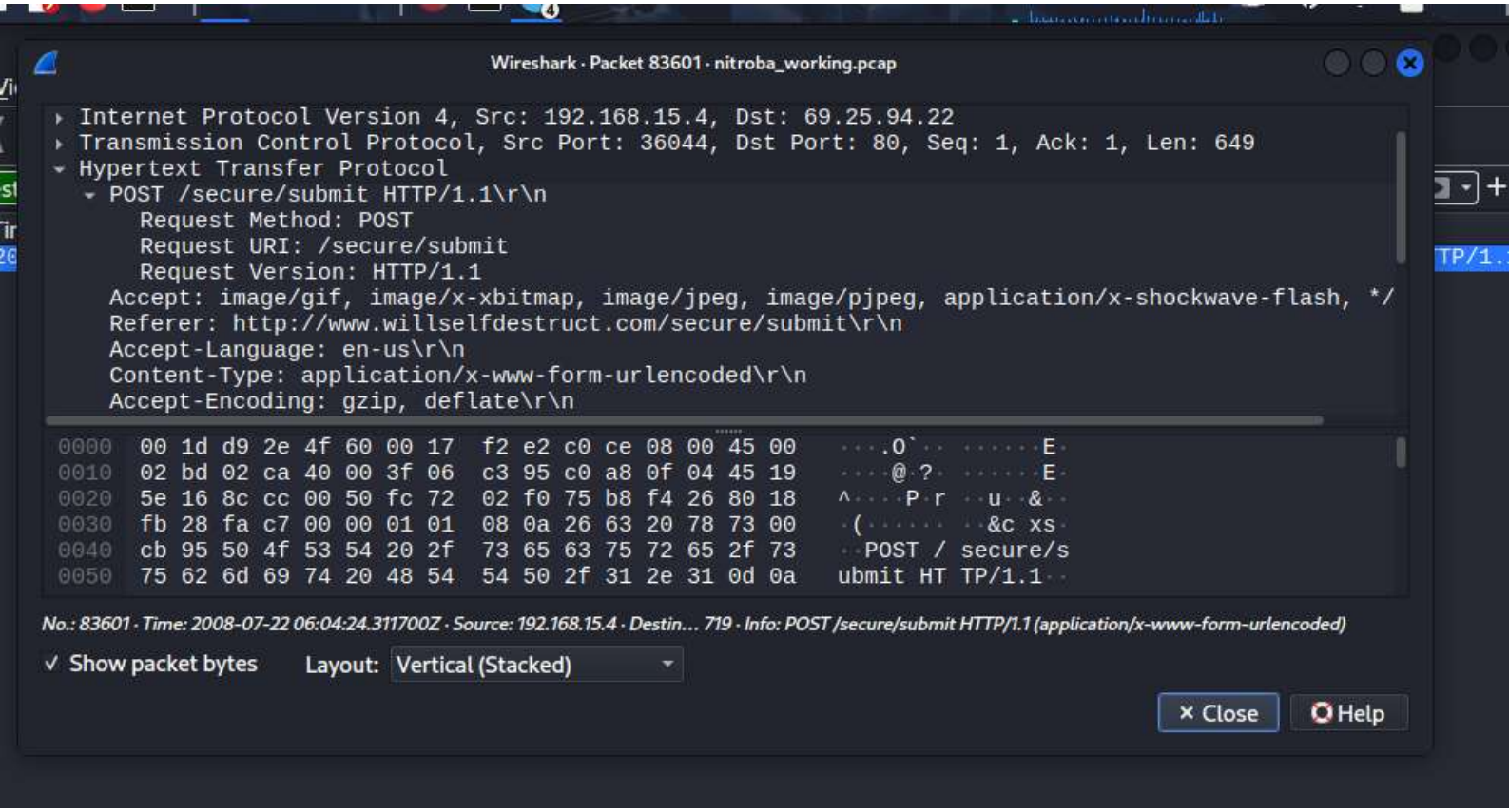

nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.1.64 74.125.19.83 TCP 70 42760 → 80 [FIN, ACK] Se
2 0.008450 74.125.19.83 192.168.1.64 TCP 70 80 → 42760 [FIN, ACK] Se
3 0.019619 192.168.1.64 74.125.19.19 HTTP 1421 GET /mail/?logout&hl=en
4 0.044170 74.125.19.19 192.168.1.64 TCP 70 80 → 35011 [ACK] Seq=1 A
5 0.224402 74.125.19.19 192.168.1.64 HTTP 1284 HTTP/1.1 302 Moved Tempo
6 0.226712 192.168.1.64 74.125.19.19 TCP 70 35011 → 80 [ACK] Seq=135
7 0.231239 192.168.1.64 74.125.19.19 TCP 70 35011 → 80 [FIN, ACK] Se
8 0.240276 74.125.19.19 192.168.1.64 TCP 70 80 → 35011 [FIN, ACK] Se
9 0.280893 192.168.1.64 192.168.1.254 DNS 78 Standard query 0x61ce A
10 0.283114 192.168.1.64 192.168.1.254 DNS 78 Standard query 0x7362 AA
11 0.294021 192.168.1.254 192.168.1.64 DNS 386 Standard query response
12 0.295200 192.168.1.254 192.168.1.64 DNS 78 Standard query response
13 0.308778 192.168.1.64 192.168.1.254 DNS 80 Standard query 0xf689 AA
14 0.321240 192.168.1.254 192.168.1.64 DNS 80 Standard query response
15 0.328385 192.168.1.64 74.125.19.103 TCP 82 39153 → 443 [SYN] Seq=0
16 0.337092 74.125.19.103 192.168.1.64 TCP 78 443 → 39153 [SYN, ACK] S
17 0.339453 74.125.19.83 192.168.1.64 TCP 70 [TCP Retransmission] 80
18 0.340872 192.168.1.64 74.125.19.103 TCP 70 39153 → 443 [ACK] Seq=1
19 0.341468 192.168.1.64 74.125.19.83 TCP 70 42760 → 80 [ACK] Seq=2 A
20 0.343355 192.168.1.64 74.125.19.103 SSLv2 172 Client Hello
21 0.353591 74.125.19.103 192.168.1.64 TCP 70 443 → 39153 [ACK] Seq=1
22 0.360576 74.125.19.103 192.168.1.64 TLSv1 1466 Server Hello
23 0.361210 74.125.19.103 192.168.1.64 TLSv1 392 Certificate, Server Hell
24 0.369537 192.168.1.64 74.125.19.103 TCP 70 39153 → 443 [ACK] Seq=10
25 0.439294 192.168.1.64 74.125.19.103 TLSv1 209 Client Key Exchange
26 0.439505 192.168.1.64 74.125.19.103 TLSv1 76 Change Cipher Spec
tal.Err

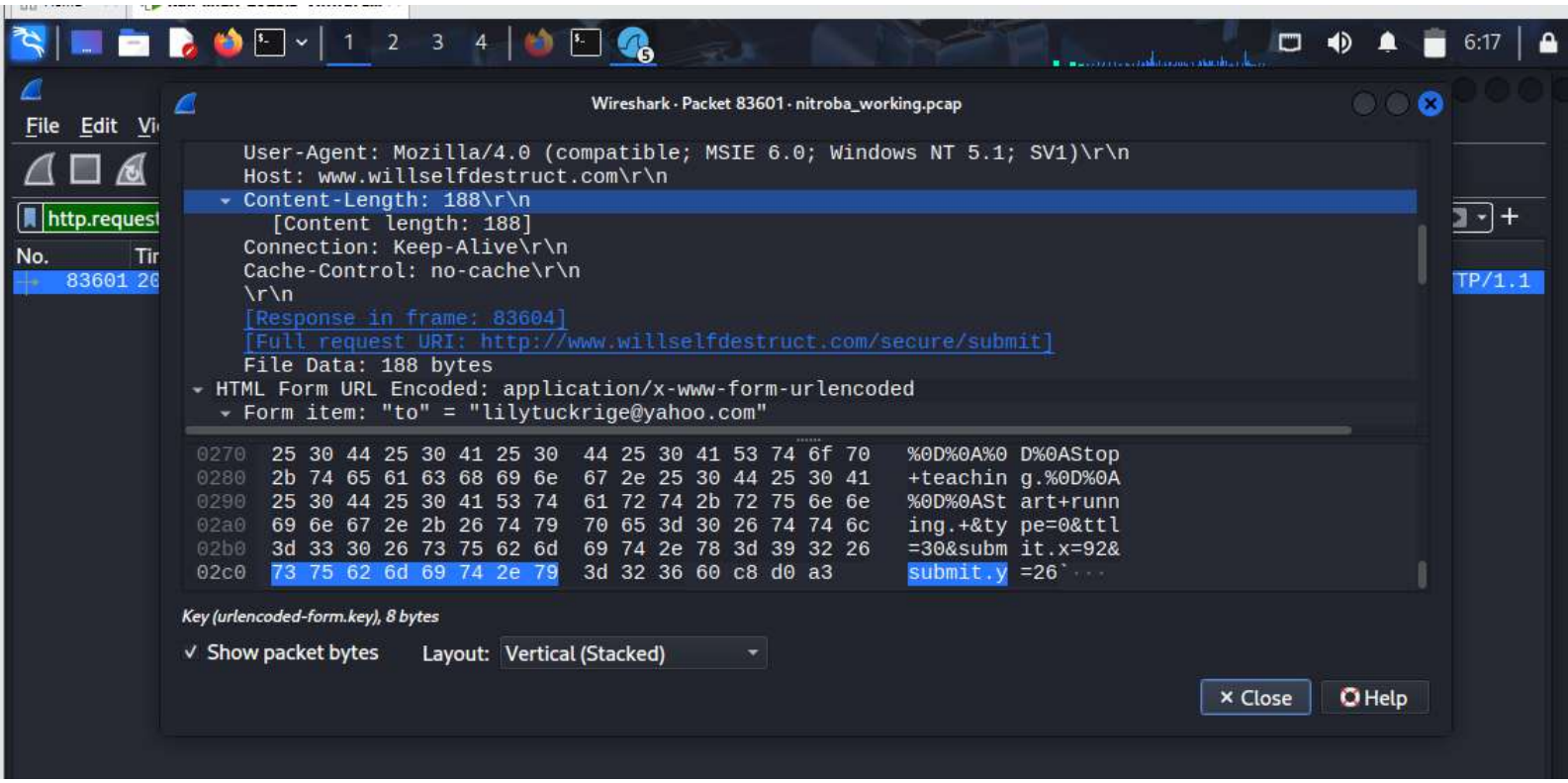

nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 2008-07-22 01:51:07.095278Z 192.168.1.64 74.125.19.83 TCP 70 42760 → 80
2 2008-07-22 01:51:07.103728Z 74.125.19.83 192.168.1.64 TCP 70 80 → 42760
3 2008-07-22 01:51:07.114897Z 192.168.1.64 74.125.19.19 HTTP 1421 GET /mail/
4 2008-07-22 01:51:07.139448Z 74.125.19.19 192.168.1.64 TCP 70 80 → 35011
5 2008-07-22 01:51:07.319680Z 74.125.19.19 192.168.1.64 HTTP 1284 HTTP/1.1 3
6 2008-07-22 01:51:07.321990Z 192.168.1.64 74.125.19.19 TCP 70 35011 → 80
7 2008-07-22 01:51:07.326517Z 192.168.1.64 74.125.19.19 TCP 70 35011 → 80
8 2008-07-22 01:51:07.335554Z 74.125.19.19 192.168.1.64 TCP 70 80 → 35011
9 2008-07-22 01:51:07.376171Z 192.168.1.64 192.168.1.254 DNS 78 Standard q
10 2008-07-22 01:51:07.378392Z 192.168.1.64 192.168.1.254 DNS 78 Standard q
11 2008-07-22 01:51:07.389299Z 192.168.1.254 192.168.1.64 DNS 386 Standard q
12 2008-07-22 01:51:07.390478Z 192.168.1.254 192.168.1.64 DNS 78 Standard q
13 2008-07-22 01:51:07.404056Z 192.168.1.64 192.168.1.254 DNS 80 Standard q
14 2008-07-22 01:51:07.416518Z 192.168.1.254 192.168.1.64 DNS 80 Standard q
15 2008-07-22 01:51:07.423663Z 192.168.1.64 74.125.19.103 TCP 82 39153 → 44
16 2008-07-22 01:51:07.432370Z 74.125.19.103 192.168.1.64 TCP 78 443 → 3915
17 2008-07-22 01:51:07.434731Z 74.125.19.83 192.168.1.64 TCP 70 [TCP Retra
18 2008-07-22 01:51:07.436150Z 192.168.1.64 74.125.19.103 TCP 70 39153 → 44
19 2008-07-22 01:51:07.436746Z 192.168.1.64 74.125.19.83 TCP 70 42760 → 80
20 2008-07-22 01:51:07.438633Z 192.168.1.64 74.125.19.103 SSLv2 172 Client Hel
21 2008-07-22 01:51:07.448869Z 74.125.19.103 192.168.1.64 TCP 70 443 → 3915
22 2008-07-22 01:51:07.455854Z 74.125.19.103 192.168.1.64 TLSv1 1466 Server Hel
23 2008-07-22 01:51:07.456488Z 74.125.19.103 192.168.1.64 TLSv1 392 Certificat
24 2008-07-22 01:51:07.464815Z 192.168.1.64 74.125.19.103 TCP 70 39153 → 44
25 2008-07-22 01:51:07.534572Z 192.168.1.64 74.125.19.103 TLSv1 209 Client Key
26 2008-07-22 01:51:07.534783Z 192.168.1.64 74.125.19.103 TLSv1 76 Change Cip

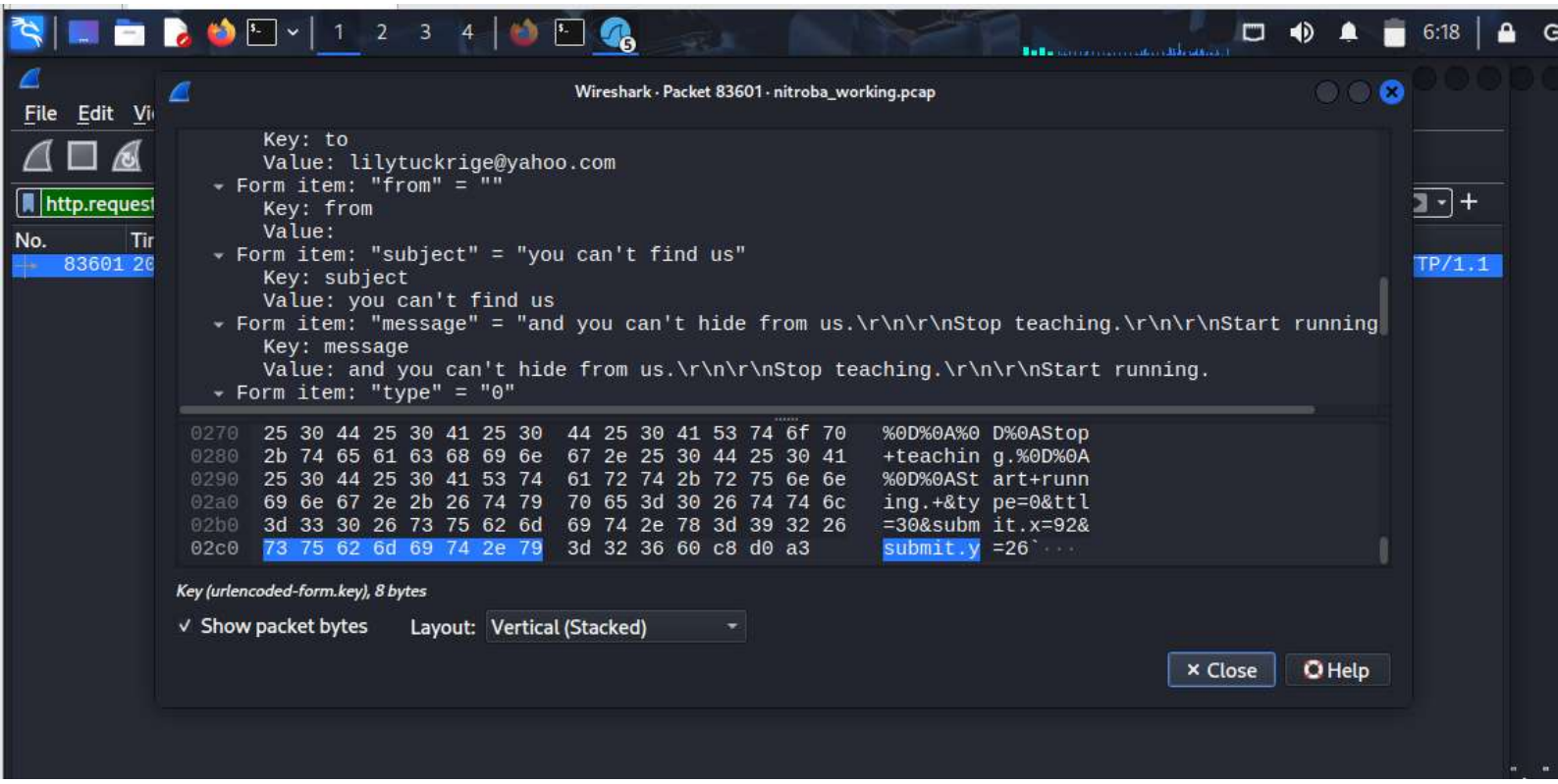

nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method == "GET" && http.host == "www.willselfdestruct.com"
No. Time Source Destination Protocol Length Info
82936 2008-07-22 06:03:43.825871Z 192.168.15.4 69.25.94.22 HTTP 596 GET /secure/submit HTTP/1.1
82985 2008-07-22 06:03:44.125949Z 192.168.15.4 69.25.94.22 HTTP 355 GET /images/spacer.gif HTTP/1
83025 2008-07-22 06:03:44.334254Z 192.168.15.4 69.25.94.22 HTTP 356 GET /images/sm-logo.gif HTTP/
83037 2008-07-22 06:03:44.374583Z 192.168.15.4 69.25.94.22 HTTP 361 GET /images/warning-home.gif
83072 2008-07-22 06:03:44.530352Z 192.168.15.4 69.25.94.22 HTTP 356 GET /images/body-bk.gif HTTP/
83087 2008-07-22 06:03:44.599079Z 192.168.15.4 69.25.94.22 HTTP 358 GET /images/bttn-send.gif HTT
83162 2008-07-22 06:03:44.852265Z 192.168.15.4 69.25.94.22 HTTP 361 GET /images/bridge_small.gif
83614 2008-07-22 06:04:24.564165Z 192.168.15.4 69.25.94.22 HTTP 461 GET /secure/success HTTP/1.1
83654 2008-07-22 06:04:24.902157Z 192.168.15.4 69.25.94.22 HTTP 360 GET /images/bk-message.gif HT

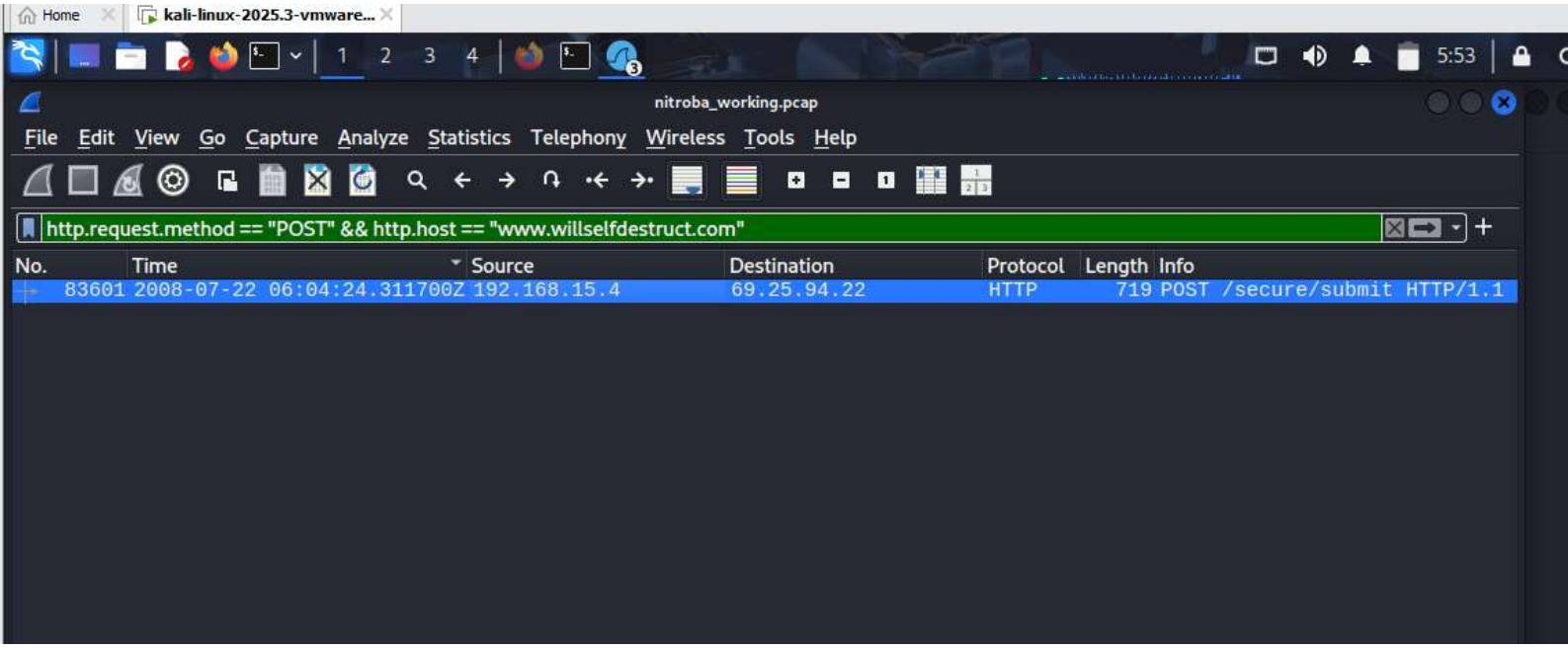

Frame 82936: Packet, 596 bytes on wire (4768 bits), 596 bytes captured (4768 bits)
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
Transmission Control Protocol, Src Port: 35984, Dst Port: 80, Seq: 1, Ack: 1, Len: 526
Hypertext Transfer Protocol
GET /secure/submit HTTP/1.1\r\n
Request Method: GET
Request URI: /secure/submit
Request Version: HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */
Referer: http://email.about.com/gi/dynamic/offsite.htm?zi=1/XJ&sdn=email&cdn=compute&tm=17&gps
Accept-Language: en-us\r\n
0000 00 1d d9 2e 4f 60 00 17 f2 e2 c0 ce 08 00 45 00
0010 02 42 5b ba 40 00 3f 06 6b 20 c0 a8 0f 04 45 19
0020 5e 16 8c 90 00 50 b4 d2 22 bc a3 e4 bb 2b 80 18
0030 fb 28 6e 7a 00 00 01 01 08 0a 26 63 1e e4 73 00
0040 2d 65 47 45 54 20 2f 73 65 63 75 72 65 2f 73 75
0050 62 6d 69 74 20 48 54 54 50 2f 31 2e 31 0d 0a 41
No.: 82936 · Time: 2008-07-22 06:03:43.825871Z · Source: 192.168.15.4 · Destinat... 69.25.94.22 · Protocol: HTTP · Length: 596 · Info: GET /secure/submit HTTP/1.1
Show packet bytes
Layout: Vertical (Stacked)
Close
Help

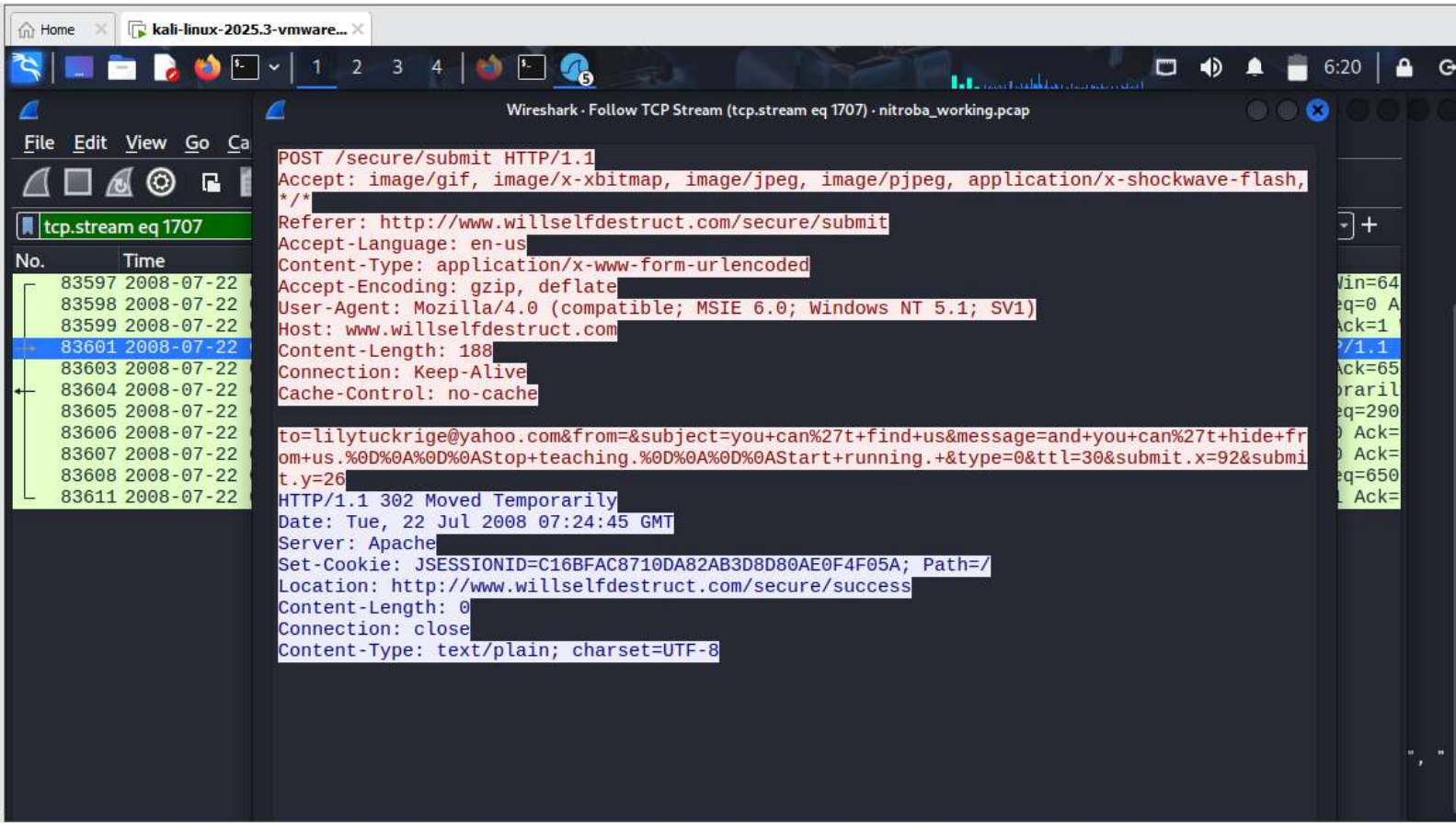

Wireshark · Packet 82936 · nitroba_working.pcap
Request Method: GET
Request URI: /secure/submit
Request Version: HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */
Referer: http://email.about.com/gi/dynamic/offsite.htm?zi=1/XJ&sdn=email&cdn=compute&tm=17&gps
Accept-Language: en-us\r\n
Accept-Encoding: gzip, deflate\r\n
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
Host: www.willselfdestruct.com\r\n
Connection: Keep-Alive\r\n
\r\n
[Response in frame: 82998]
0000 00 1d d9 2e 4f 60 00 17 f2 e2 c0 ce 08 00 45 00
0010 02 42 5b ba 40 00 3f 06 6b 20 c0 a8 0f 04 45 19
0020 5e 16 8c 90 00 50 b4 d2 22 bc a3 e4 bb 2b 80 18
0030 fb 28 6e 7a 00 00 01 01 08 0a 26 63 1e e4 73 00
0040 2d 65 47 45 54 20 2f 73 65 63 75 72 65 2f 73 75
0050 62 6d 69 74 20 48 54 54 50 2f 31 2e 31 0d 0a 41
No.: 82936 · Time: 2008-07-22 06:03:43.825871Z · Source: 192.168.15.4 · Destinat... 69.25.94.22 · Protocol: HTTP · Length: 596 · Info: GET /secure/submit HTTP/1.1
Show packet bytes
Layout: Vertical (Stacked)
Close
Help

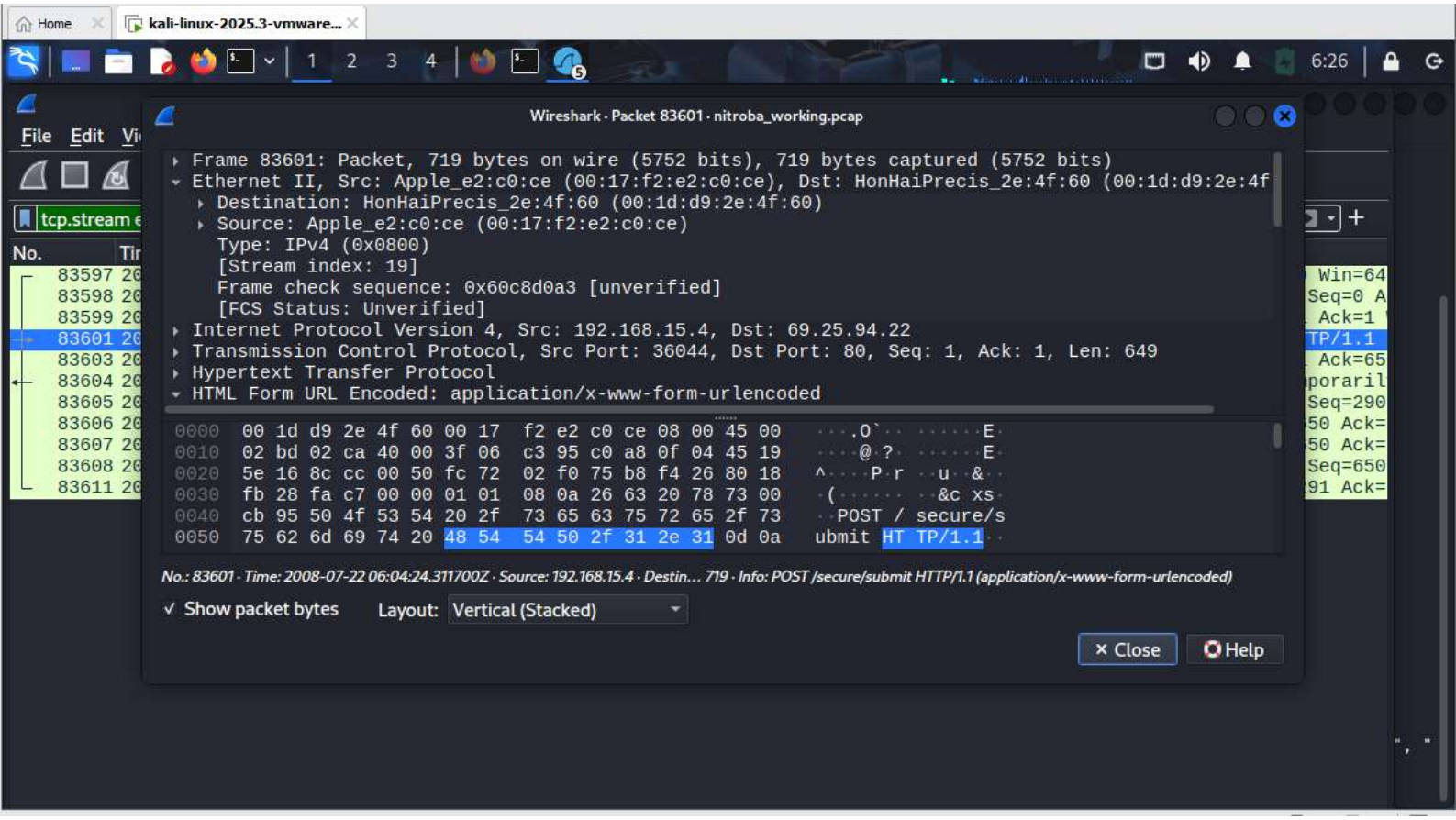

Wireshark · Packet 83601 · nitroba_working.pcap
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
Transmission Control Protocol, Src Port: 36044, Dst Port: 80, Seq: 1, Ack: 1, Len: 649
Hypertext Transfer Protocol
POST /secure/submit HTTP/1.1\r\n
Request Method: POST
Request URI: /secure/submit
Request Version: HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */
Referer: http://www.willselfdestruct.com/secure/submit\r\n
Accept-Language: en-us\r\n
Content-Type: application/x-www-form-urlencoded\r\n
Accept-Encoding: gzip, deflate\r\n
0000 00 1d d9 2e 4f 60 00 17 f2 e2 c0 ce 08 00 45 00
0010 02 bd 02 ca 40 00 3f 06 c3 95 c0 a8 0f 04 45 19
0020 5e 16 8c cc 00 50 fc 72 02 f0 75 b8 f4 26 80 18
0030 fb 28 fa c7 00 00 01 01 08 0a 26 63 20 78 73 00
0040 cb 95 50 4f 53 54 20 2f 73 65 63 75 72 65 2f 73
0050 75 62 6d 69 74 20 48 54 54 50 2f 31 2e 31 0d 0a
No.: 83601 · Time: 2008-07-22 06:04:24.311700Z · Source: 192.168.15.4 · Destin... 719 · Info: POST /secure/submit HTTP/1.1 (application/x-www-form-urlencoded)
Show packet bytes
Layout: Vertical (Stacked)
Close
Help

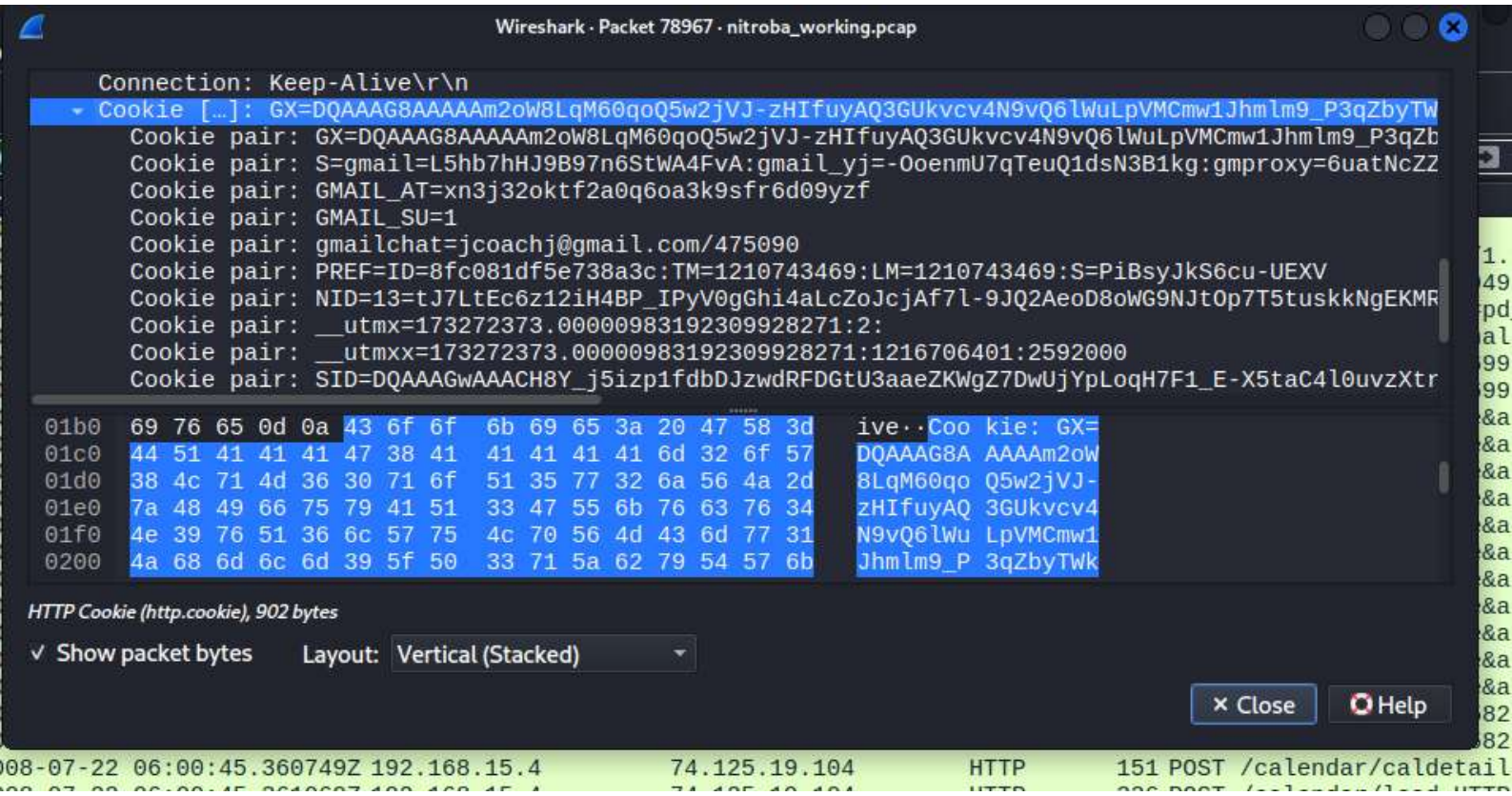

Wireshark · Packet 83601 · nitroba_working.pcap
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)\r\n
Host: www.willselfdestruct.com\r\n
Content-Length: 188\r\n
[Content length: 188]
Connection: Keep-Alive\r\n
Cache-Control: no-cache\r\n
\r\n
[Response in frame: 83604]
[Full request URI: http://www.willselfdestruct.com/secure/submit]
File Data: 188 bytes
HTML Form URL Encoded: application/x-www-form-urlencoded
Form item: "to" = "lilytuckrige@yahoo.com"
0270 25 30 44 25 30 41 25 30 44 25 30 41 53 74 6f 70 %0D%0A%0 D%0AStop
0280 2b 74 65 61 63 68 69 6e 67 2e 25 30 44 25 30 41 +teachin g.%0D%0A
0290 25 30 44 25 30 41 53 74 61 72 74 2b 72 75 6e 6e %0D%0ASt art+runn
02a0 69 6e 67 2e 2b 26 74 79 70 65 3d 30 26 74 74 6c ing.+&ty pe=0&ttl
02b0 3d 33 30 26 73 75 62 6d 69 74 2e 78 3d 39 32 26 =30&subm it.x=92&
02c0 73 75 62 6d 69 74 2e 79 3d 32 36 60 c8 d0 a3 submit.y =26`
Key (urlencoded-form.key), 8 bytes
Show packet bytes
Layout: Vertical (Stacked)
Close
Help


Wireshark · Packet 83601 · nitroba_working.pcap
Key: to
Value: lilytuckrige@yahoo.com
Form item: "from" = ""
Key: from
Value:
Form item: "subject" = "you can't find us"
Key: subject
Value: you can't find us
Form item: "message" = "and you can't hide from us.\r\n\r\nStop teaching.\r\n\r\nStart running
Key: message
Value: and you can't hide from us.\r\n\r\nStop teaching.\r\n\r\nStart running.
Form item: "type" = "0"
0270 25 30 44 25 30 41 25 30 44 25 30 41 53 74 6f 70 %0D%0A%0 D%0AStop
0280 2b 74 65 61 63 68 69 6e 67 2e 25 30 44 25 30 41 +teachin g.%0D%0A
0290 25 30 44 25 30 41 53 74 61 72 74 2b 72 75 6e 6e %0D%0ASt art+runn
02a0 69 6e 67 2e 2b 26 74 79 70 65 3d 30 26 74 74 6c ing.+&ty pe=0&ttl
02b0 3d 33 30 26 73 75 62 6d 69 74 2e 78 3d 39 32 26 =30&subm it.x=92&
02c0 73 75 62 6d 69 74 2e 79 3d 32 36 60 c8 d0 a3 submit.y =26`...
Key (urlencoded-form.key), 8 bytes
Show packet bytes
Layout: Vertical (Stacked)
Close
Help

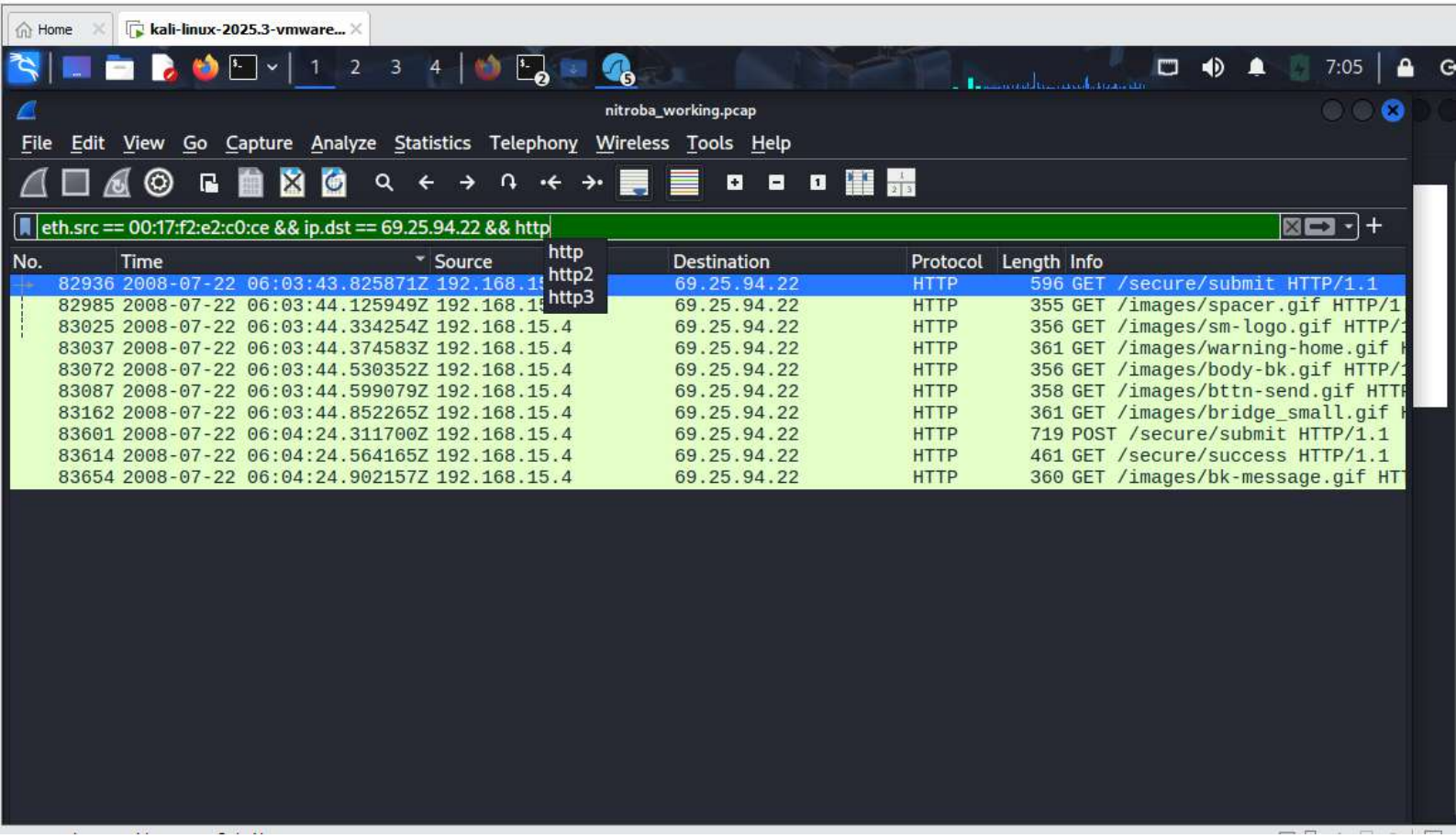

Home
kali-linux-2025.3-vmware...
1 2 3 4
5:53
nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method == "POST" && http.host == "www.willselfdestruct.com"
No. Time Source Destination Protocol Length Info
83601 2008-07-22 06:04:24.311700Z 192.168.15.4 69.25.94.22 HTTP 719 POST /secure/submit HTTP/1.1

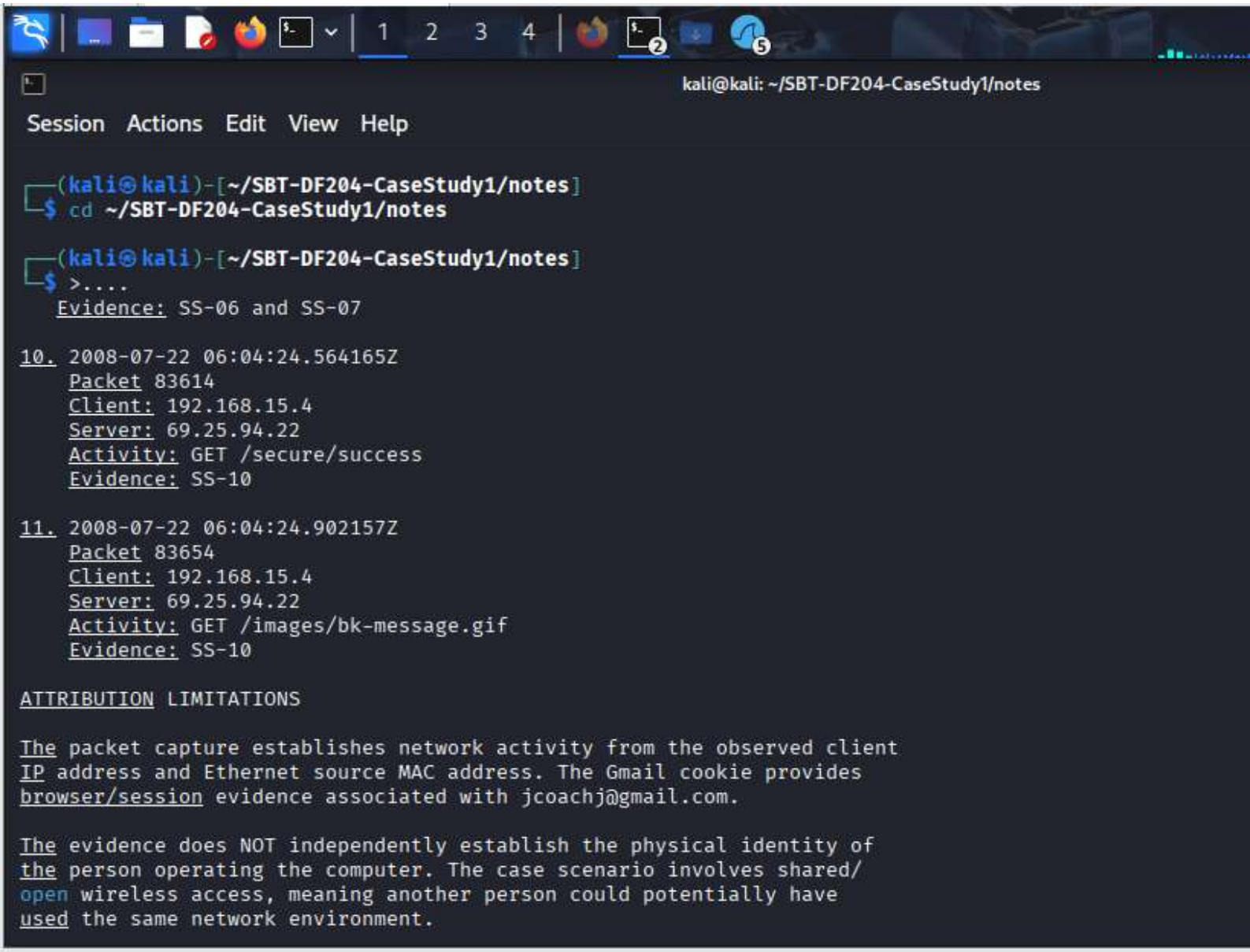

Home
kali-linux-2025.3-vmware...
1 2 3 4
6:20
Wireshark · Follow TCP Stream (tcp.stream eq 1707) · nitroba_working.pcap
File Edit View Go Ca
tcp.stream eq 1707
No. Time
83597 2008-07-22
83598 2008-07-22
83599 2008-07-22
83601 2008-07-22
83603 2008-07-22
83604 2008-07-22
83605 2008-07-22
83606 2008-07-22
83607 2008-07-22
83608 2008-07-22
83611 2008-07-22
POST /secure/submit HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */*
Referer: http://www.willselfdestruct.com/secure/submit
Accept-Language: en-us
Content-Type: application/x-www-form-urlencoded
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)
Host: www.willselfdestruct.com
Content-Length: 188
Connection: Keep-Alive
Cache-Control: no-cache
to=lilytuckrige@yahoo.com&from=&subject=you+can%27t+find+us&message=and+you+can%27t+hide+from+us.%0D%0A%0D%0AStop+teaching.%0D%0A%0D%0AStart+running.+&type=0&ttl=30&submit.x=92&submit.y=26
HTTP/1.1 302 Moved Temporarily
Date: Tue, 22 Jul 2008 07:24:45 GMT
Server: Apache
Set-Cookie: JSESSIONID=C16BFAC8710DA82AB3D8D80AE0F4F05A; Path=/
Location: http://www.willselfdestruct.com/secure/success
Content-Length: 0
Connection: close
Content-Type: text/plain; charset=UTF-8


Home
kali-linux-2025.3-vmware...
6:26
File Edit
tcp.stream
No.
Wireshark · Packet 83601 · nitroba_working.pcap
Frame 83601: Packet, 719 bytes on wire (5752 bits), 719 bytes captured (5752 bits)
Ethernet II, Src: Apple_e2:c0:ce (00:17:f2:e2:c0:ce), Dst: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f
Destination: HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)
Source: Apple_e2:c0:ce (00:17:f2:e2:c0:ce)
Type: IPv4 (0x0800)
[Stream index: 19]
Frame check sequence: 0x60c8d0a3 [unverified]
[FCS Status: Unverified]
Internet Protocol Version 4, Src: 192.168.15.4, Dst: 69.25.94.22
Transmission Control Protocol, Src Port: 36044, Dst Port: 80, Seq: 1, Ack: 1, Len: 649
Hypertext Transfer Protocol
HTML Form URL Encoded: application/x-www-form-urlencoded
0000 00 1d d9 2e 4f 60 00 17 f2 e2 c0 ce 08 00 45 00
0010 02 bd 02 ca 40 00 3f 06 c3 95 c0 a8 0f 04 45 19
0020 5e 16 8c cc 00 50 fc 72 02 f0 75 b8 f4 26 80 18
0030 fb 28 fa c7 00 00 01 01 08 0a 26 63 20 78 73 00
0040 cb 95 50 4f 53 54 20 2f 73 65 63 75 72 65 2f 73
0050 75 62 6d 69 74 20 48 54 54 50 2f 31 2e 31 0d 0a
ubmit HT TP/1.1
No.: 83601 · Time: 2008-07-22 06:04:24.311700Z · Source: 192.168.15.4 · Destin... 719 · Info: POST /secure/submit HTTP/1.1 (application/x-www-form-urlencoded)
Show packet bytes
Layout: Vertical (Stacked)
Close
Help



008-07-22 06:00:45.360749Z 192.168.15.4 74.125.19.104 HTTP 151 POST /calendar/caldetail


Home
kali-linux-2025.3-vmware...
1 2 3 4
7:05
nitroba_working.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
eth.src == 00:17:f2:e2:c0:ce && ip.dst == 69.25.94.22 && http
http
http2
http3
No. Time Source Destination Protocol Length Info
82936 2008-07-22 06:03:43.825871Z 192.168.15 69.25.94.22 HTTP 596 GET /secure/submit HTTP/1.1
82985 2008-07-22 06:03:44.125949Z 192.168.15 69.25.94.22 HTTP 355 GET /images/spacer.gif HTTP/1
83025 2008-07-22 06:03:44.334254Z 192.168.15.4 69.25.94.22 HTTP 356 GET /images/sm-logo.gif HTTP/1
83037 2008-07-22 06:03:44.374583Z 192.168.15.4 69.25.94.22 HTTP 361 GET /images/warning-home.gif H
83072 2008-07-22 06:03:44.530352Z 192.168.15.4 69.25.94.22 HTTP 356 GET /images/body-bk.gif HTTP/1
83087 2008-07-22 06:03:44.599079Z 192.168.15.4 69.25.94.22 HTTP 358 GET /images/bttn-send.gif HTTP
83162 2008-07-22 06:03:44.852265Z 192.168.15.4 69.25.94.22 HTTP 361 GET /images/bridge_small.gif H
83601 2008-07-22 06:04:24.311700Z 192.168.15.4 69.25.94.22 HTTP 719 POST /secure/submit HTTP/1.1
83614 2008-07-22 06:04:24.564165Z 192.168.15.4 69.25.94.22 HTTP 461 GET /secure/success HTTP/1.1
83654 2008-07-22 06:04:24.902157Z 192.168.15.4 69.25.94.22 HTTP 360 GET /images/bk-message.gif HT


1 2 3 4
kali@kali: ~/SBT-DF204-CaseStudy1/notes
Session Actions Edit View Help
(kali㉿kali)-[~/SBT-DF204-CaseStudy1/notes]
$ cd ~/SBT-DF204-CaseStudy1/notes
(kali㉿kali)-[~/SBT-DF204-CaseStudy1/notes]
$ >....
Evidence: SS-06 and SS-07
10. 2008-07-22 06:04:24.564165Z
Packet 83614
Client: 192.168.15.4
Server: 69.25.94.22
Activity: GET /secure/success
Evidence: SS-10
11. 2008-07-22 06:04:24.902157Z
Packet 83654
Client: 192.168.15.4
Server: 69.25.94.22
Activity: GET /images/bk-message.gif
Evidence: SS-10
ATTRIBUTION LIMITATIONS
The packet capture establishes network activity from the observed client
IP address and Ethernet source MAC address. The Gmail cookie provides
browser/session evidence associated with jcoachj@gmail.com.
The evidence does NOT independently establish the physical identity of
the person operating the computer. The case scenario involves shared/
open wireless access, meaning another person could potentially have
used the same network environment.